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POTHOLE User's Manual

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TopAGNPS v6.00.a.000

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POTHOLE User's Manual:

Introduction:

“POTHOLE” is a module in the suite of modules called TopAGNPS. Its purpose is to allow the user to identify potholes within a watershed. The user can specify a threshold area value as a selection criteria for reporting purposes.

Most of the input files for POTHOLE, as described below, can be generated by executing the following modules: DEDNM, RASPRO, RASFOR, and AgFlow. The TopAGNPS modules, (DEDNM, RASPRO, and RASFOR), produces spatial files required by POTHOLE. AgFlow produces a spatial output file called “AnnAGNPS_Reach_IDs.asc” which is also required by POTHOLE.

POTHOLE produces several output files which are described below.

Input files:

There are six input files, described below, necessary for the successful execution of POTHOLE. The first is a control file and the remaining five are spatial layer files. The layer files are specifically named as described below.

Control file:

POTHOLE.CSV or POTHOLE.XML

The required control file allows the user to specify a number of parameters that affect the processing of spatial data and the determination of potholes. If both “POTHOLE.CSV” and “POTHOLE.XML” are found, the “POTHOLE.CSV” control file will be processed. A message will be produced in the log file and POTHOLE execution terminated if the control file is not present.

“POTHOLE.CSV” is a csv-formatted control file. The first record in the control file is the header record containing four possible keywords that may be used. The second record is the data record containing the values for the keywords that are specified in the header record. Only two records are read, header and data; therefore, any other information contained within the control file is ignored. The control file must be named “POTHOLE.CSV”.

The following is an Excel example of the csv-formatted control file showing all of the available keywords:

POTHOLE_OPTION	POTHOLE_SURFACE_AREA	ASC_PATH	CSV_PATH
0	0.025	./Output_Files/ASC	./Output_Files/CSV

“POTHOLE.XML” is xml-formatted control file. If the xml-format is used, it must adhere strictly to the following –only one tag per line, with no spaces. The xml-formatted control file must be named “POTHOLE.XML”. The following is an example of the xml-formatted control file showing all available keywords.

```
<TOPAGNPS>
<KEYWORD ID="ASC_PATH">./Output_Files/ASC</KEYWORD>
<KEYWORD ID="CSV_PATH">./Output_Files/CSV</KEYWORD>
<KEYWORD ID="POTHOLE_OPTION">0</KEYWORD>
<KEYWORD ID="POTHOLE_SURFACE_AREA">0.025</KEYWORD>
</TOPAGNPS>
```

Keywords:

All parameters may be present but only required parameters and any desired optional parameters need be present. There are four possible keywords in the control file. All keywords must be spelled exactly as specified below and have a comma separating the keywords. The keywords are case and order insensitive.

“POTHOLE_OPTION”:

This optional keyword allows the user to designate if potholes will be identified and processed. If this parameter is not present in the control file or if the value for this parameter is blank then the default value is "0" meaning that potholes will be identified.

“POTHOLE_SURFACE_AREA”:

This optional keyword allows the user to designate the minimum area in hectares that is required for a pothole to be considered valid and processed for reporting purposes; that is, the minimum surface area threshold has been met.

“ASC_Path”:

This optional keyword allows the user to specify the path to where the spatial files created by DEDNM, RASPRO, RASFOR, and AgFlow are located. This path will also be used for spatial output files created by POTHOLE.

This keyword is optional and if not included then the default location of the spatial input files and POTHOLE spatial output files will be assumed. The default relative path is the folder where TopAGNPS was invoked. If the keyword is present with a path specified then that path will be used for all of the required spatial input files and POTHOLE spatial output files. If the keyword is

used but its value is blank, then the current path from which TopAGNPS is executed will be used for all of the spatial input files and POTHOLE spatial output files.

NOTE: This keyword is only used for independent runs of the POTHOLE module. It is ignored when POTHOLE is run in sequence with DEDNM, RASPRO, RASFOR, and AgFlow. In this case the output files from POTHOLE will be placed in the default folder location.

“CSV_Path”:

This optional keyword allows the user to specify the path to where the csv-formatted output files created by POTHOLE will be written.

This keyword is optional and if not included then the default location of the POTHOLE csv-formatted output files will be assumed. The default relative path is the folder where TopAGNPS was invoked. If the keyword is present with a path specified then that path will be used for all of the POTHOLE csv-formatted output files. If the keyword is used but its value is blank, then the current path from which TopAGNPS is executed will be used for all of the POTHOLE csv-formatted output files.

NOTE: This keyword is only used for independent runs of the POTHOLE module. It is ignored when POTHOLE is run in sequence with DEDNM, RASPRO, RASFOR, and AgFlow. In this case the output files from POTHOLE will be placed in the default folder location.

Input data files:

If executing TopAGNPS on a UNIX/Linux system (case sensitive), the following list is in the expected case for each file.

The following input data files described below are spatial layers. The spatial layers must have the same dimensions with the same header information in each file. All of the filenames must be as specified below.

BOUND.ASC (TopAGNPS):

This required spatial input file is the watershed boundary layer.

DEPFLT.ASC (TopAGNPS):

This required spatial input file is the depression/flat area layer.

FLOVEC.ASC (TopAGNPS):

This required spatial input file is the flow vector layer.

UPAREA.ASC (TopAGNPS):

This required spatial input file is the drainage area layer.

AnnAGNPS_Reach_IDs.asc (AgFlow):

This required spatial input file is the reach (AGNPS reach) designations layer.

Output files:

There are currently nine output files generated by the program.

Log file:

The log file, "POTHOLE.log", is automatically generated and is a log of the processing as the program continues its execution. Additionally, the log files for "TopAGNPS" will also contain information specific for POTHOLE. The log file, as it is being written, is also displayed to the screen. The log file contains processing information as well as error and warning messages.

Output data files:

AnnAGNPS_Pothole_Data_Section.csv:

This output file is a csv-formatted file that contains a list of the potholes. Potholes are currently viewed as wetlands and can be imported as the "Wetland Data" section into the AGNPS Input Editor or used directly by AnnAGNPS if used with the AnnAGNPS csv input option.

Pothole_Depressions_All_Codes_Within_DEM.asc:

This is a spatial layer file containing all identified depression/flat areas within the input DEM as defined by DEDNM and contains data values "0" through "6" where "0" is the no-data value. Definitions for "1" through "6" can be found in the "TOPAZ_User_Manual.pdf" documentation. The POTHOLE module only processes values "1" and "2".

Pothole_Depressions_Codes_1_and_2_Within_Watershed.asc:

This is a spatial layer file containing all identified depression/flat areas coded with “1” and “2” within the watershed boundary. Definitions of codes “1” and “2” can be found in the “TOPAZ_User_Manual.pdf” documentation. The POTHOLE module only processes values “1” and “2”.

Pothole_Elevations_Within_Depressions.asc:

This is a spatial layer file containing elevation data for all identified depression/flat areas coded with “1” and “2” within the watershed boundary.

Pothole_IDs_All_Within_Watershed.asc:

This is a spatial layer file containing assigned IDs for all potholes within the watershed boundary.

Pothole_IDs_Meeting_Criteria.asc:

This is a spatial layer file containing assigned IDs for all potholes within the watershed boundary that met the minimum surface area criteria.

Pothole_IDs_Meeting_Criteria.csv:

This is a csv-formatted file containing information for all potholes within the watershed boundary that met the minimum surface area criteria.

Pothole_IDs_Not_Meeting_Criteria.csv:

This is a csv-formatted file containing information for all potholes within the watershed boundary that did not meet the minimum surface area criteria including a reason code.